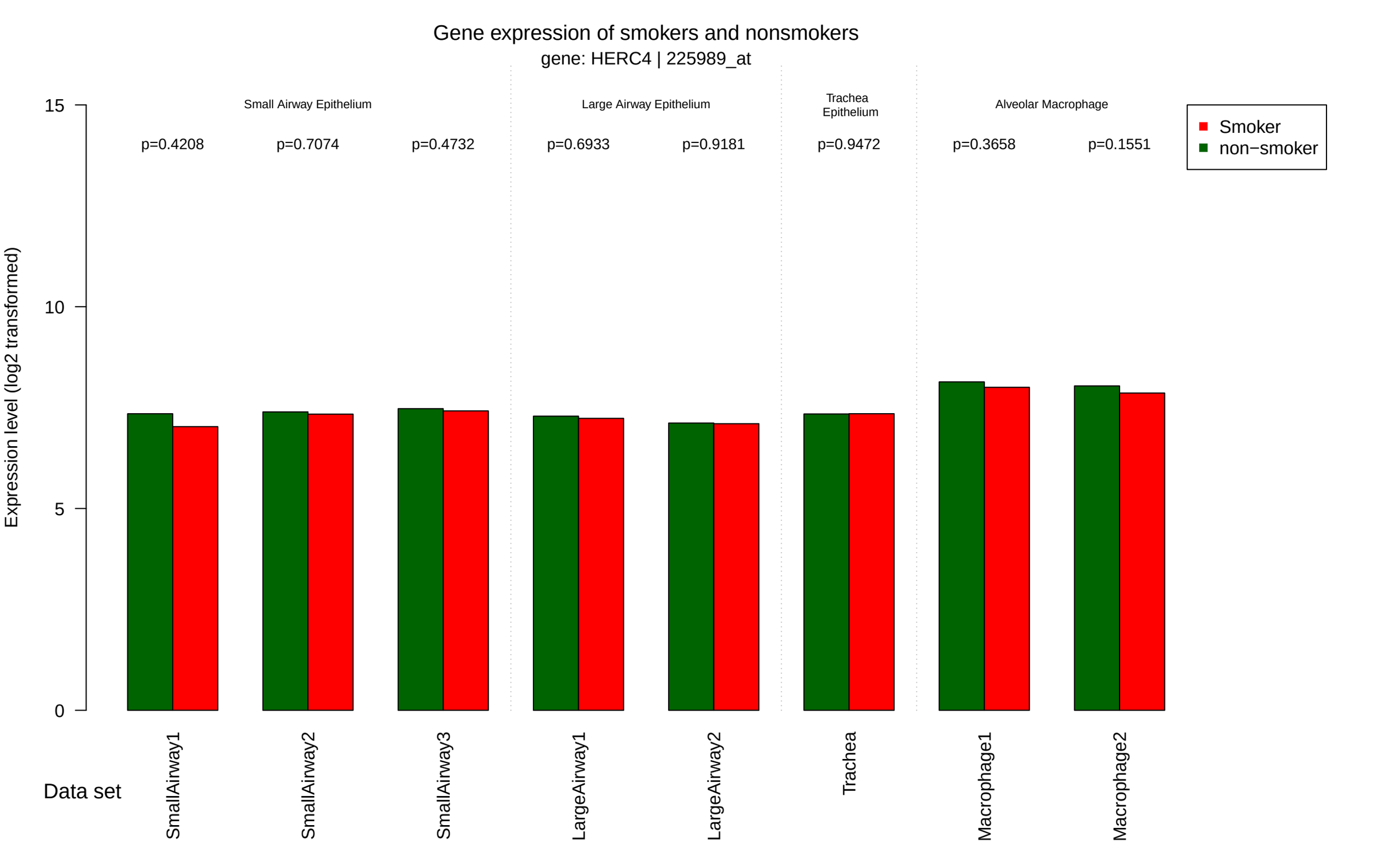
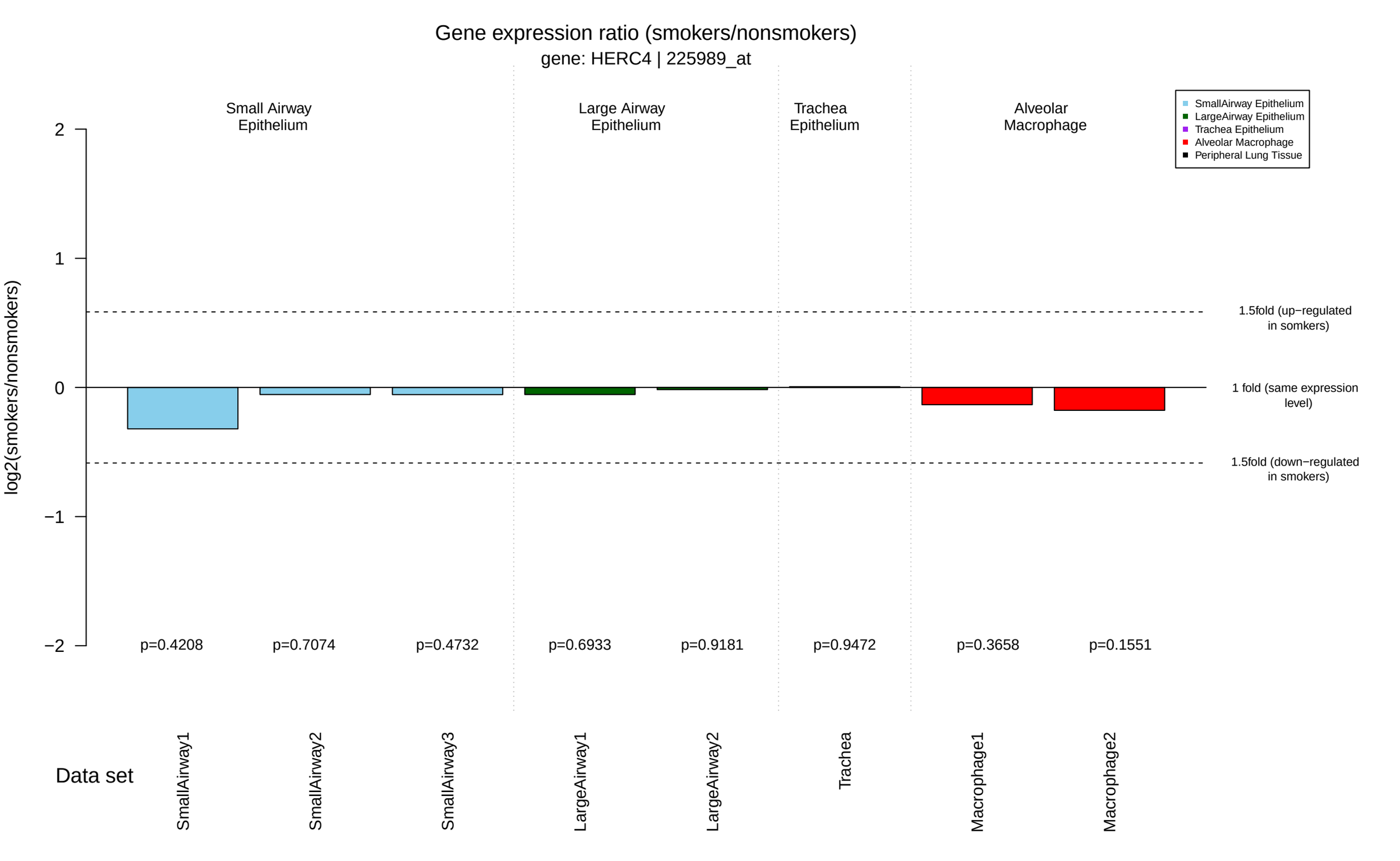
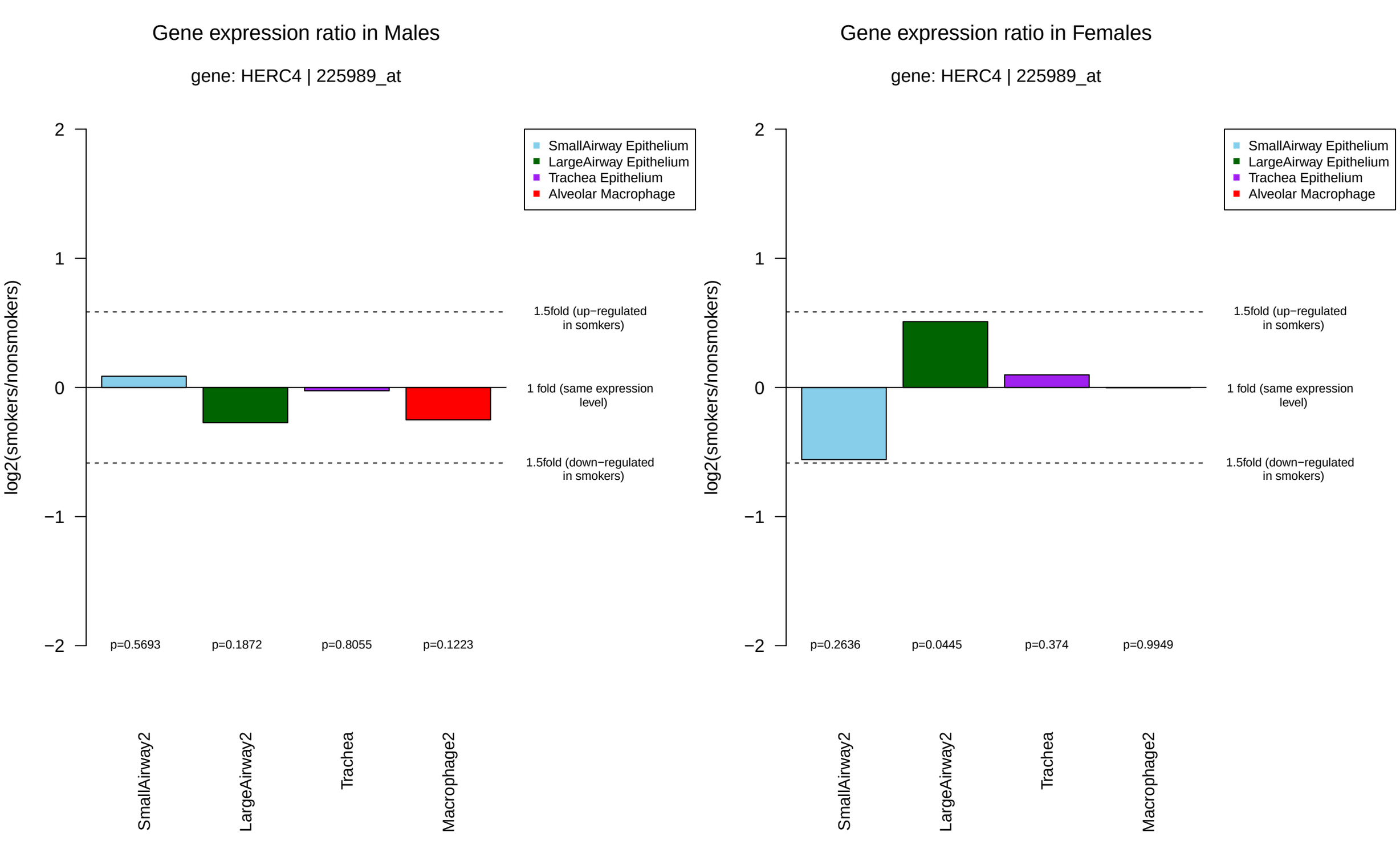
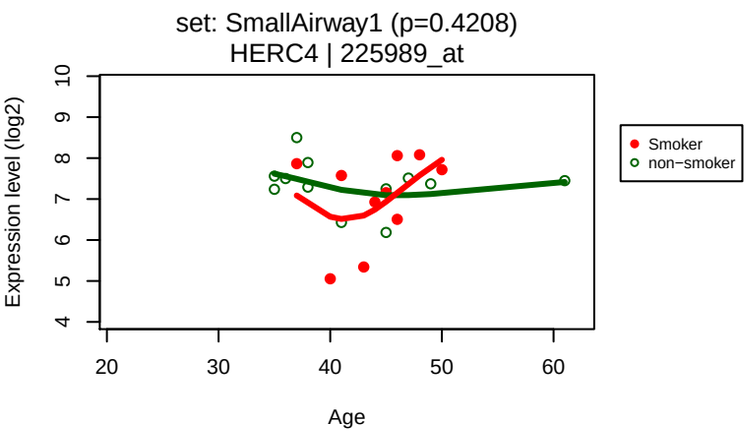
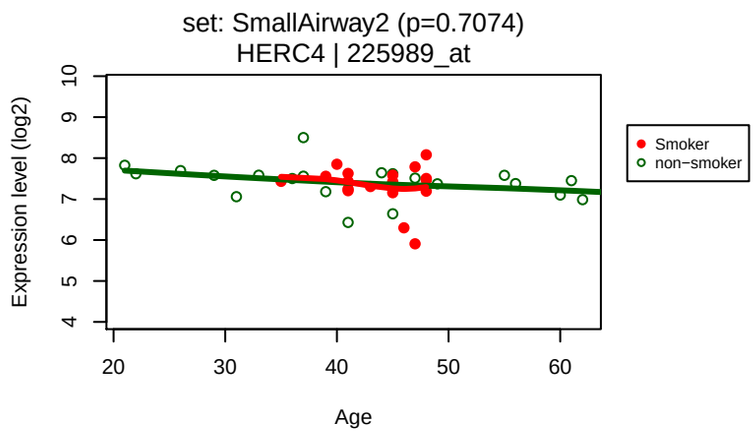
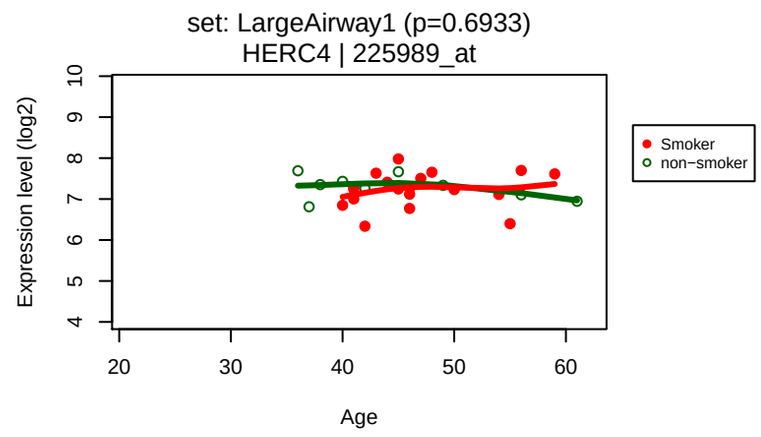

Gene expression of smokers and nonsmokers
gene: HERC4 | 225989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4208
p=0.7074
p=0.4732
p=0.6933
p=0.9181
p=0.9472
p=0.3658
p=0.1551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

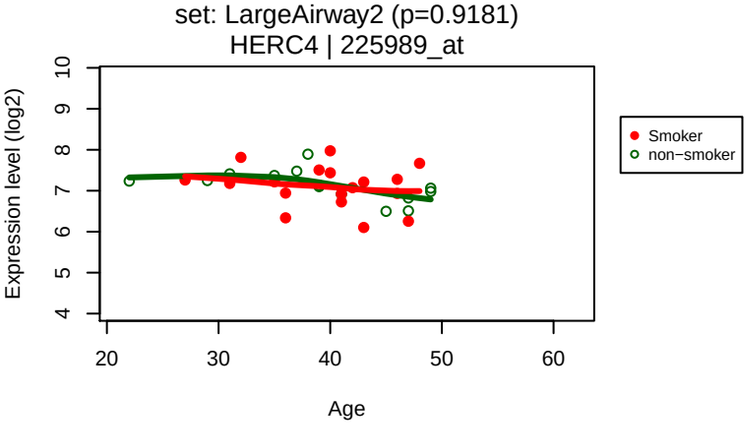

Gene expression ratio (smokers/nonsmokers)
gene: HERC4 | 225989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4208
p=0.7074
p=0.4732
p=0.6933
p=0.9181
p=0.9472
p=0.3658
p=0.1551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

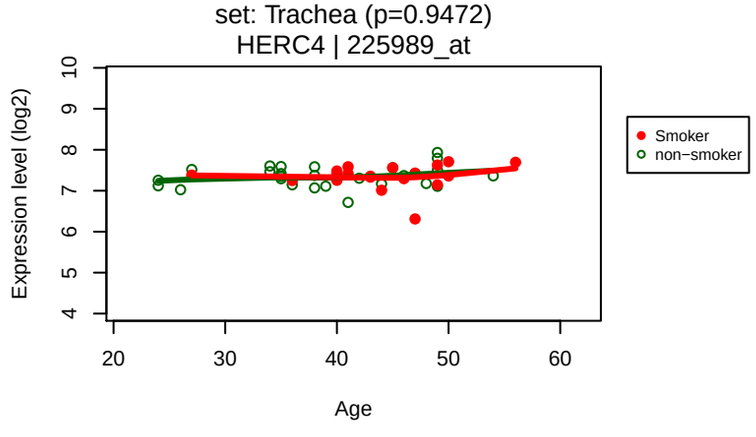

Gene expression ratio in Males
gene: HERC4 | 225989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5693
p=0.1872
p=0.8055
p=0.1223
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC4 | 225989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2636
p=0.0445
p=0.374
p=0.9949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

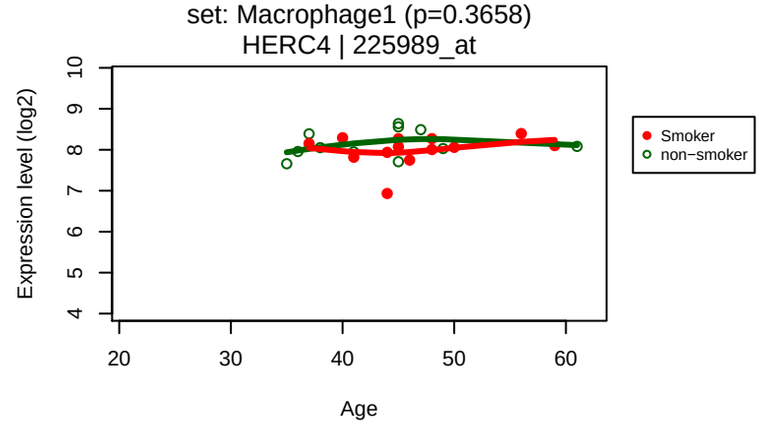

set: SmallAirway1 (p=0.4208)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker

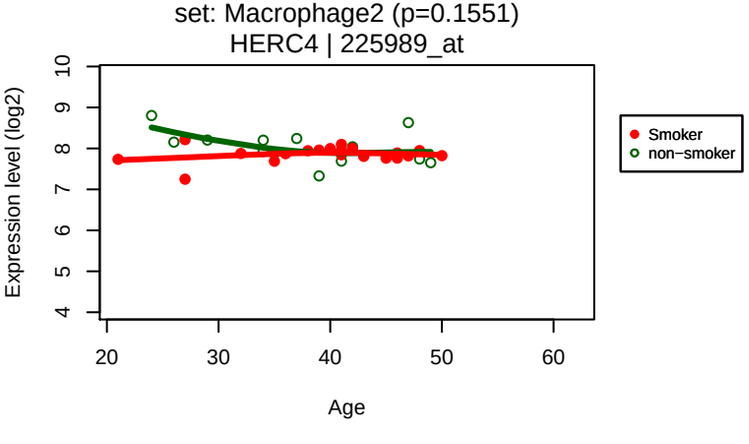

set: SmallAirway2 (p=0.7074)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker

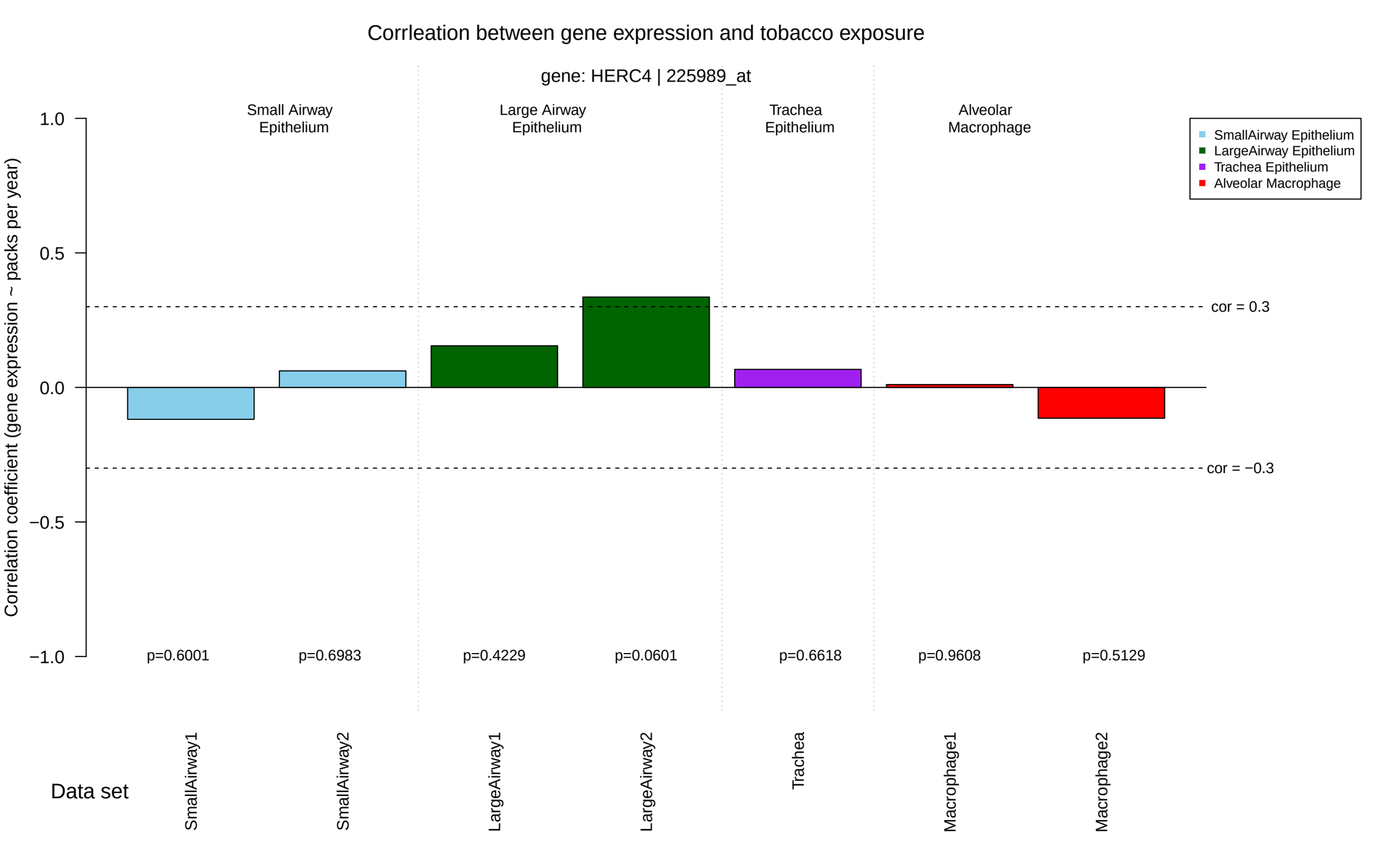

set: LargeAirway1 (p=0.6933)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9181)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9472)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3658)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1551)
HERC4 | 225989_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC4 | 225989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6001
p=0.6983
p=0.4229
p=0.0601
p=0.6618
p=0.9608
p=0.5129
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2